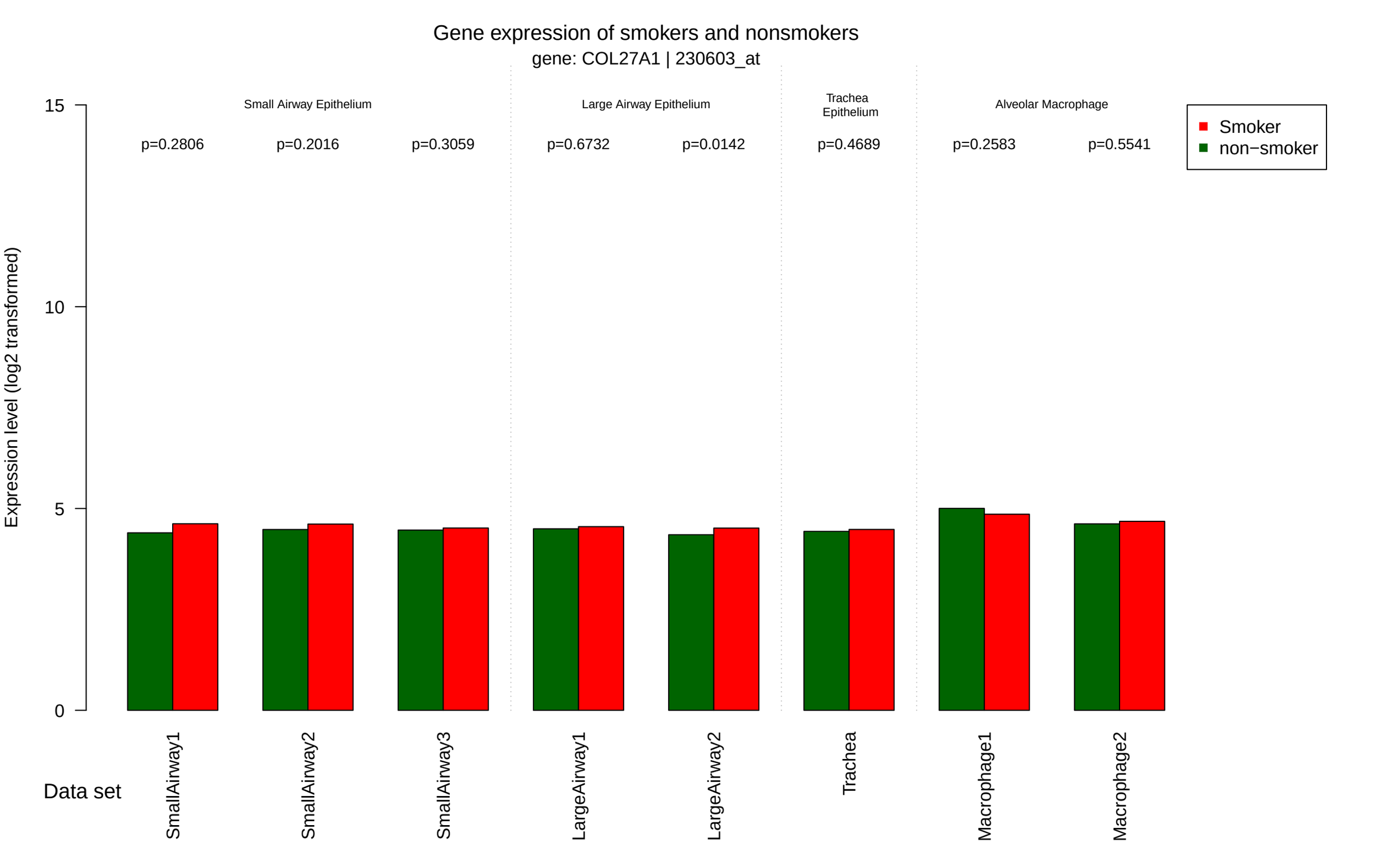
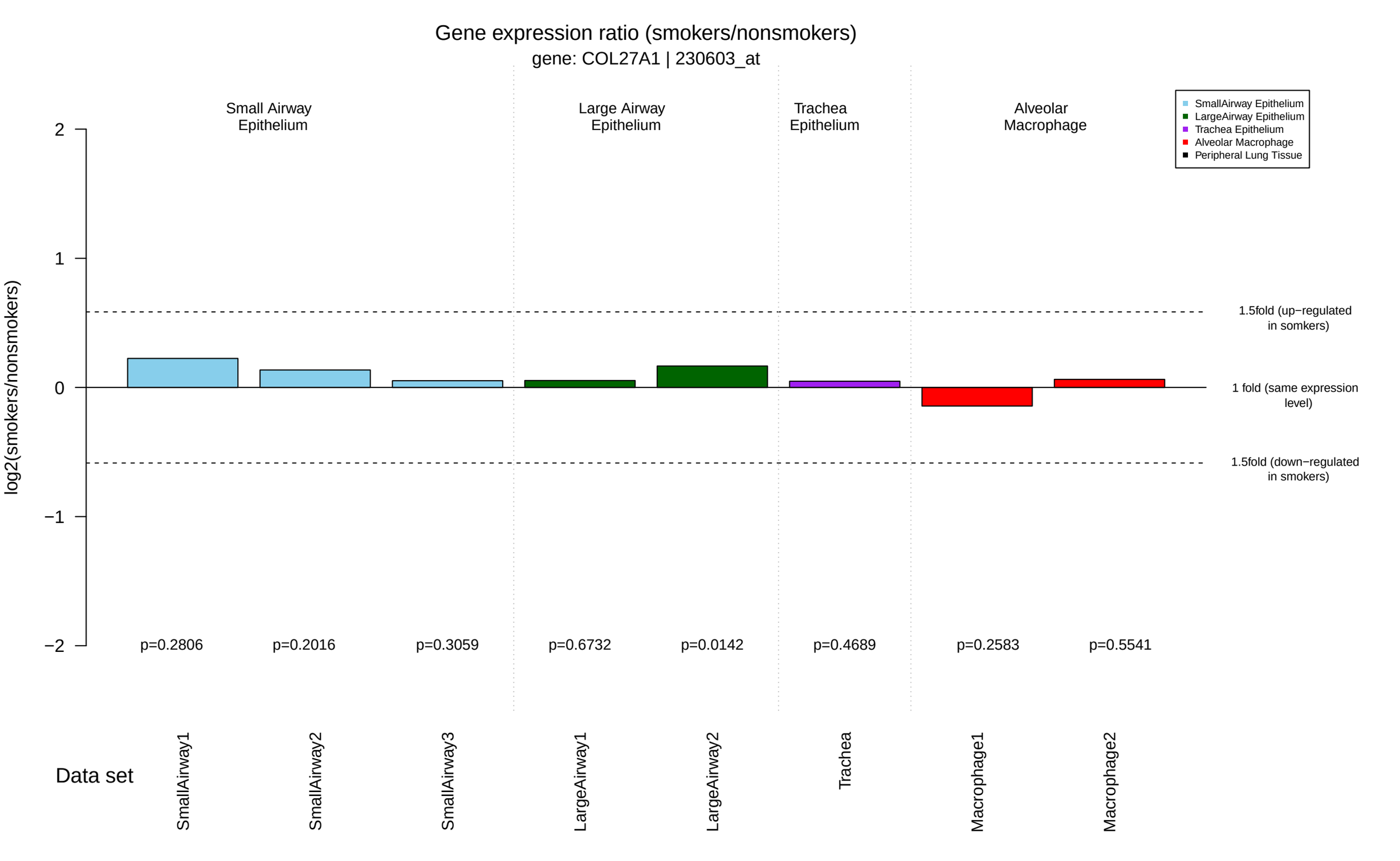
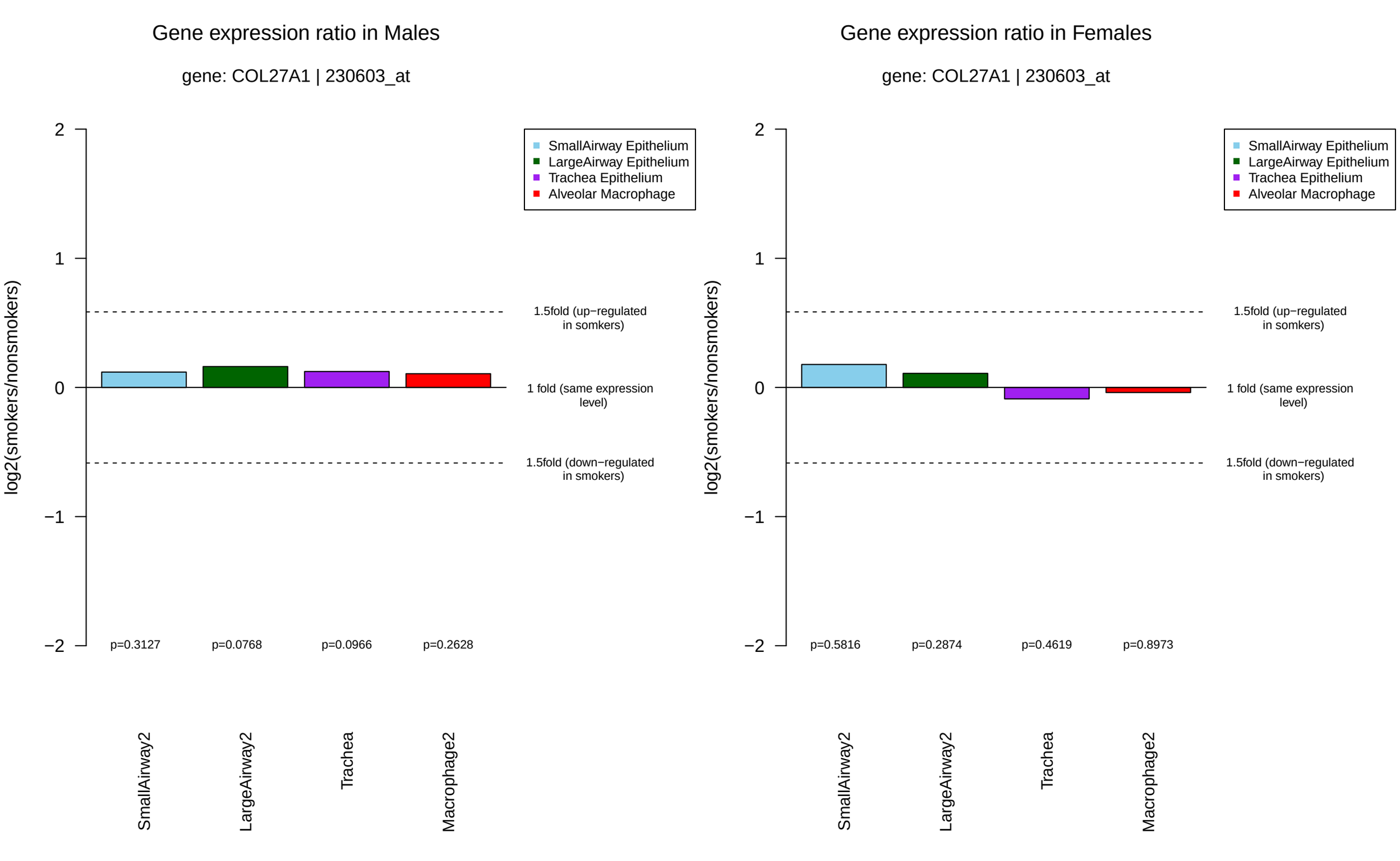
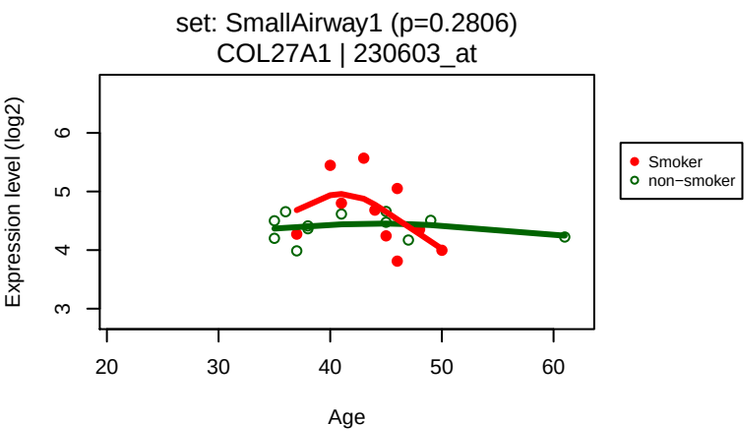
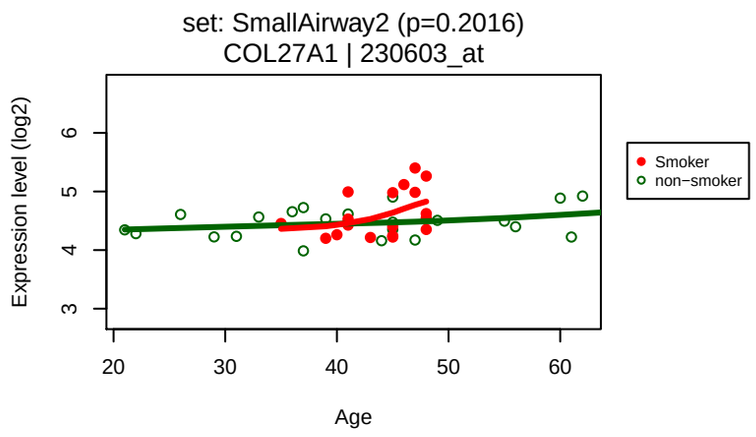
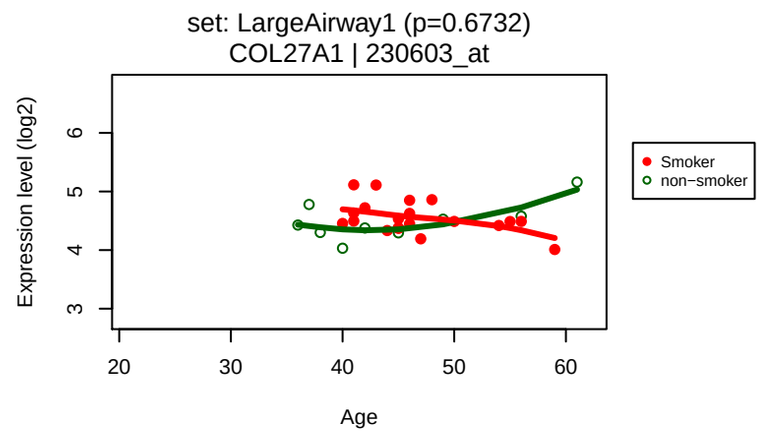

Gene expression of smokers and nonsmokers
gene: COL27A1 | 230603_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2806
p=0.2016
p=0.3059
p=0.6732
p=0.0142
p=0.4689
p=0.2583
p=0.5541
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

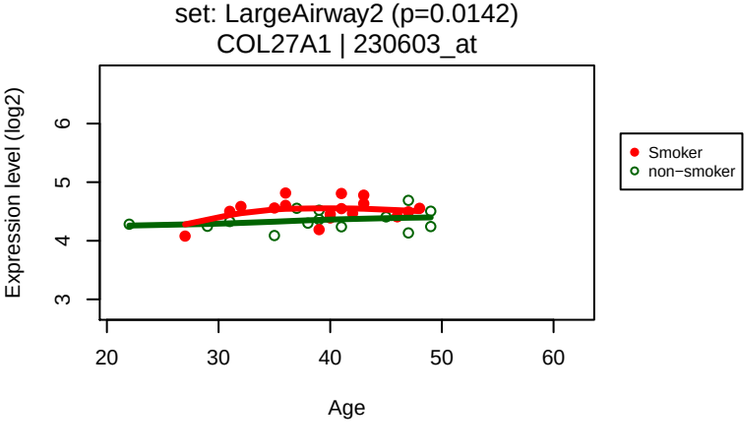

Gene expression ratio (smokers/nonsmokers)
gene: COL27A1 | 230603_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2806
p=0.2016
p=0.3059
p=0.6732
p=0.0142
p=0.4689
p=0.2583
p=0.5541
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

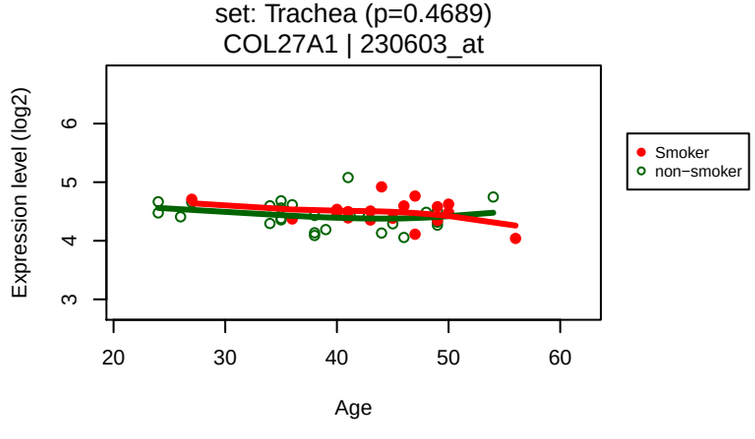

Gene expression ratio in Males
gene: COL27A1 | 230603_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3127
p=0.0768
p=0.0966
p=0.2628
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL27A1 | 230603_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5816
p=0.2874
p=0.4619
p=0.8973
SmallAirway2
LargeAirway2
Trachea
Macrophage2

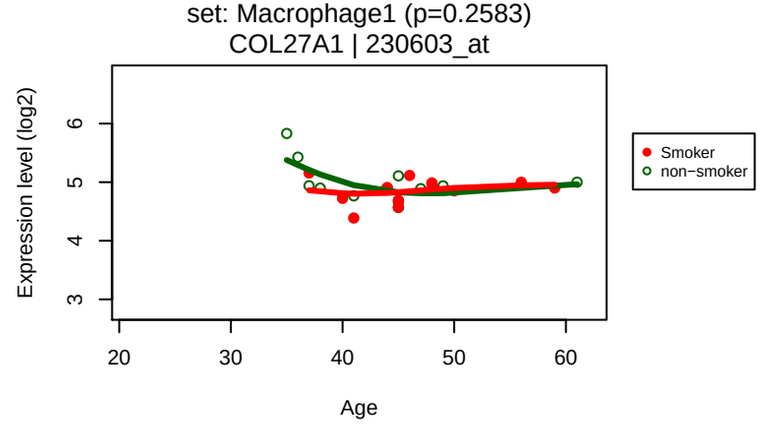

set: SmallAirway1 (p=0.2806)
COL27A1 | 230603_at
Expression level (log2)
Age
Smoker
non−smoker

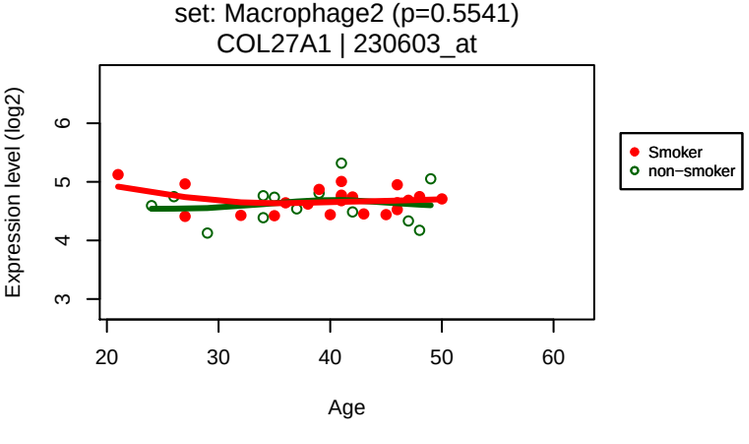

set: SmallAirway2 (p=0.2016)
COL27A1 | 230603_at
Expression level (log2)
Age
Smoker
non−smoker

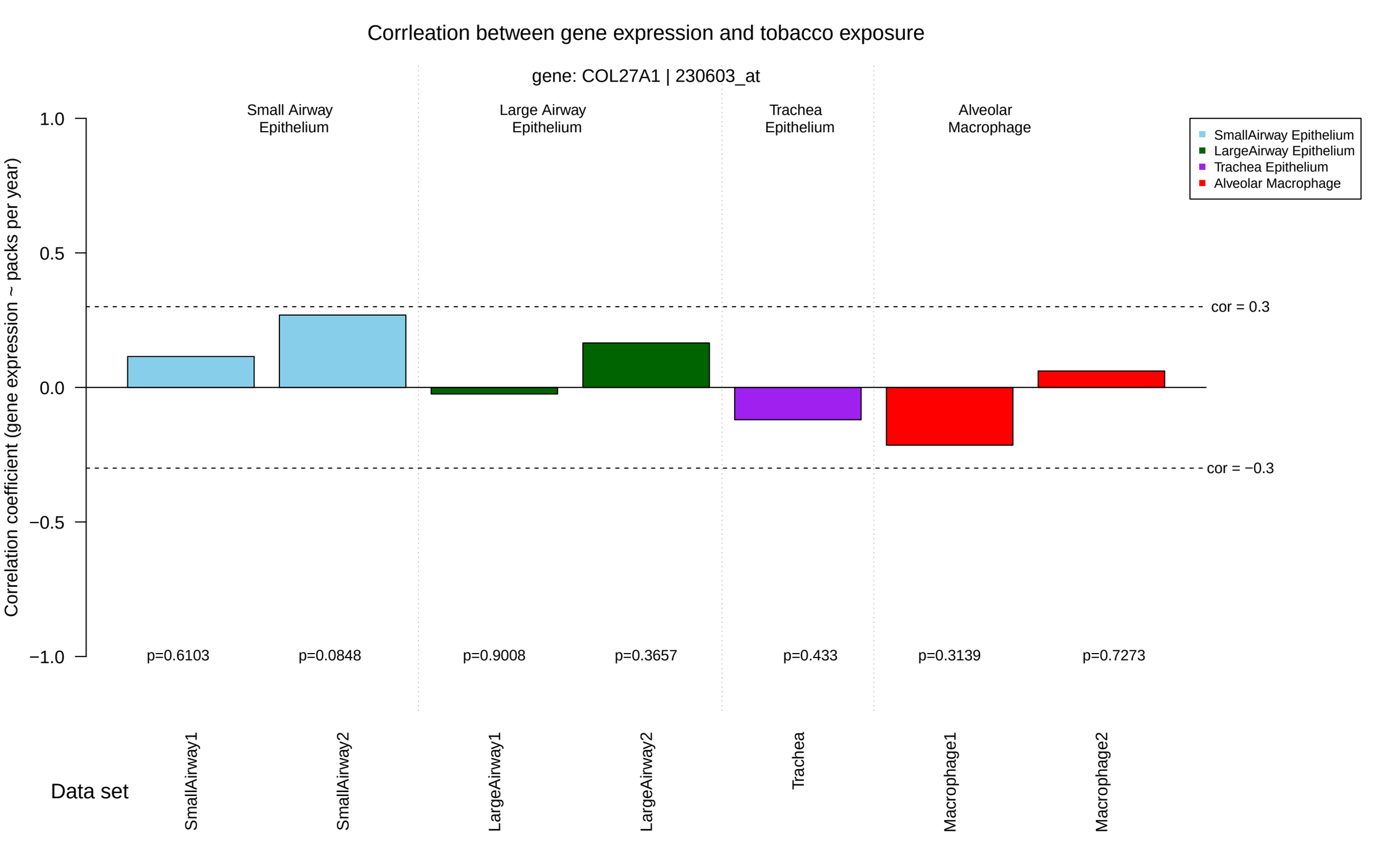

set: LargeAirway1 (p=0.6732)
COL27A1 | 230603_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0142)
COL27A1 | 230603_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4689)
COL27A1 | 230603_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2583)
COL27A1 | 230603_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5541)
COL27A1 | 230603_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL27A1 | 230603_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6103
p=0.0848
p=0.9008
p=0.3657
p=0.433
p=0.3139
p=0.7273
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2